



FPT UNIVERSITY

CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

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I. Record of Changes

Date	A* M, D	In charge	Change Description
16/05/2026	A	LocPTCE171503	Added Section 1: Overview
16/05/2026	A	LamTTCE181752	Added Section 2: Product Background
16/05/2026	A	KhangKCCE180324	Added Section 3: Existing Systems
16/05/2026	A	HoangLHCE181003	Added Section 4: Business Opportunity
16/05/2026	A	HoangLHCE181003	Added Section 5: Software Product Vision
16/05/2026	A	LocPTCE171503	Added Section 6.1: Major Features
16/05/2026	A	HauNDCE180785	Added Section 6.2: Limitations & Exclusions
17/05/2026	M	HauNDCE180785	Updated Section 6.1: Major Features
17/05/2026	M	KhangKCCE180324	Revised document formatting and layout
27/05/2026	M	LamTTCE181752	Updated Section 5: Software Product Vision
27/05/2026	M	LamTTCE181752	Updated Section 2.4: Practical and Socio-Economic Values
28/05/2026	M	LocPTCE171503	Updated Figure 9: Major Features diagram
11/08/2026	M	HauNDCE180785	Revised Sections 2.3–2.7 to align the platform background, technologies, authentication, and realtime communication descriptions with the current system.
11/08/2026	M	HoangLHCE181003	Updated Section 4: Business Opportunity to clarify the marketplace model, target Clients, and value proposition of the platform.
11/08/2026	M	LamTTCE181752	Updated Section 5: Software Product Vision to align the marketplace, SaaS

			management, shift operations, and service communication with the current project scope.
11/08/2026	M	LocPTCE171503	Updated Section 6.1: Major Features and revised Figure 9 to reflect the current feature scope and system workflows.
11/08/2026	M	KhangKCCE180324	Updated Section 6.2: Limitations & Exclusions to clarify attendance validation, browser permissions, location usage, and other system limitations.
04/09/2026	M	HauNDCE180785	Updated Section 6.1: Reorganized into 12 major features, included all 6 user roles, and fully synchronized with Figure 9.
04/09/2026	M	HoangLHCE181003	Updated Section 6.2: Expanded scope limitations (LI-01 to LI-08), clarifying attendance verification and external integrations.

*A - Added M - Modified D - Deleted

II. Project Introduction

1. Overview

1.1 Project Information

- Project name: SGCMP - Security Guard Connection & Management Platform
- Project code: SEP490_18
- Group name: PatrolX Team
- Software type: Website

1.2 Project Team

a. Supervisor

Full Name	Role	Email	Mobile
Lam Le Huan	Lecturer	huanll@fe.edu.vn	0913737372

b. Team Members

Full Name	Role	Email	Mobile
Nguyen Dinh Hau	Leader	HauNDCE180785@fpt.edu.vn	0852933924
Phan Tan Loc	Member	LocPTCE171503@fpt.edu.vn	0983162593
Lam Huy Hoang	Member	HoangLHCE181003@fpt.edu.vn	0968811709
Tran Thanh Lam	Member	LamTTCE181752@fpt.edu.vn	0368543865
Kim Chi Khang	Member	KhangKCCE180324@fpt.edu.vn	0385559008

2. Product Background

In the modern era, security services are an essential part of maintaining the safety and continuity of business and community activities. Security companies provide a wide range of services, from guarding offices, residential areas, factories, warehouses, and construction sites to supporting events and other locations that require professional security personnel. Unlike many conventional service businesses, security operations depend heavily on the ability to deploy the right personnel at the right place and time. A single service contract may involve multiple guards, different working shifts, specific qualifications, and continuous coordination between the company, its employees, and the customer. As a result, managing security services requires not only effective workforce administration but also efficient coordination between service providers and customers.

From the customer's perspective, finding and selecting an appropriate security service provider can also be challenging. Customers may have to search through different websites, social media channels, personal recommendations, or individual company contacts to identify potential providers. The information available from these sources is often inconsistent and difficult to compare. Customers may not have a standardized way to evaluate important criteria such as service types, pricing, service areas, company profiles, availability, and previous customer reviews. In addition, the process of requesting quotations, discussing service requirements, confirming bookings, and tracking service status may require communication through multiple channels,

which can increase the time and effort required to arrange a security service. The absence of a centralized platform that connects customers with multiple security companies therefore limits transparency and makes the service selection process less convenient.

Security companies also face significant operational challenges in managing their workforce. A security company may have a large number of guards working across multiple customer locations and different shifts. Managing recurring schedules, employee availability, leave requests, shift changes, and replacement guards manually can result in scheduling conflicts, insufficient staffing, or delays in assigning personnel. In addition, maintaining employee information and monitoring documents such as professional certificates, identification documents, and other qualifications becomes increasingly difficult as the number of employees grows. Without a centralized system, managers may also have limited visibility into the actual attendance and operational status of guards working at remote locations.

Communication and incident management represent another important operational challenge. When an incident occurs at a customer site, information may be communicated through telephone calls, messaging applications, or manual reports. Such processes can make it difficult for managers to maintain a consistent record of incidents and respond quickly. Similarly, manually recording guard check-in and check-out information can make it difficult to verify attendance accurately and monitor whether assigned personnel are present at their designated locations. These limitations indicate the need for a specialized information system that supports both workforce management and service coordination within the security service industry.

Existing general-purpose platforms can support some aspects of the problem but do not fully address the operational requirements of security service providers. Recruitment and job-matching platforms primarily focus on connecting employers with potential workers and generally do not provide comprehensive post-hiring workforce management capabilities. On the other hand, general enterprise management or ERP solutions may provide extensive business functions but often require considerable configuration and may not include workflows specifically designed for security services, such as guard assignment, shift scheduling, attendance tracking, service requests, and incident management. This creates an opportunity for a specialized platform that addresses both the customer-facing and internal operational aspects of the security service business.

To address these challenges, the Security Guard Connection & Management Platform (SGCMP) is proposed as a multi-tenant platform that integrates a security service marketplace with internal management capabilities for security companies.

3. Existing Systems

3.1 Tracktik.com

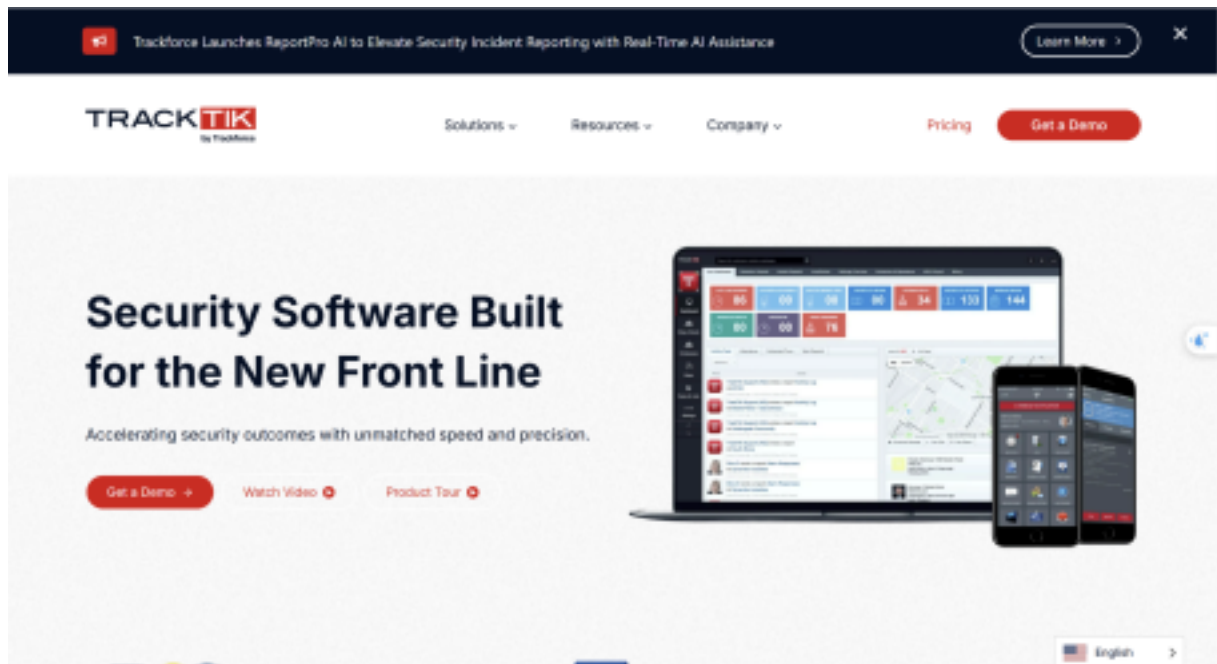


Figure 7. tracktik.com Web

TrackTik is a globally recognized, comprehensive cloud-based security workforce management software. It provides end-to-end solutions for security firms, focusing heavily on live tracking and guard touring.

Homepage: tracktik.com

- **Features:** Live GPS tracking, NFC/Barcode checkpoint scanning, scheduling, incident reporting, and billing.
- **Benefits:** Highly robust, feature-rich, and proven in the enterprise security market. Excellent mobile application for guards.
- **Drawbacks:** It is strictly a B2B operational software, not a marketplace. Clients cannot use TrackTik to find and hire security companies. Furthermore, the pricing is extremely high for small to medium-sized enterprises in developing markets.

3.2 Silvertrac Software

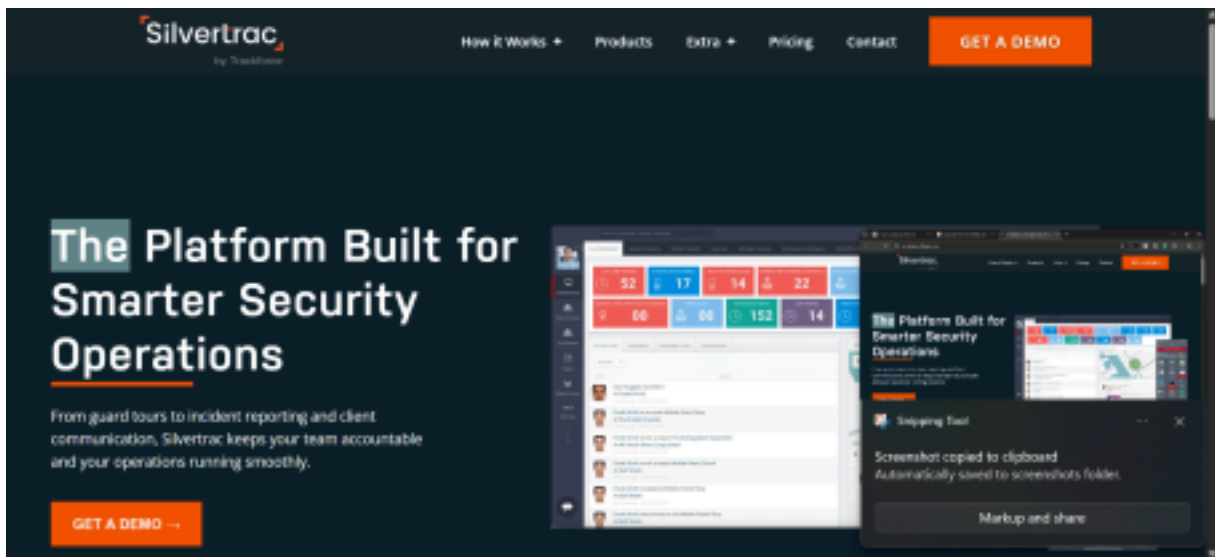


Figure 8. Silvertrac Software Web

Silvertrac is a guard management and incident reporting software aimed at improving accountability and communication between security guards and property managers.

Homepage: [Silvertrac Software](#)

- **Features:** Real-time issue reporting, guard tour tracking, automated daily activity reports.
- **Benefits:** Simple to set up and very effective for creating professional, real-time reports for clients.
- **Drawbacks:** Similar to TrackTik, it lacks the marketplace functionality. It focuses mostly on the reporting aspect rather than full-scale HR and financial contract management

4. Business Opportunity

The current market landscape in Vietnam presents a highly lucrative, untapped opportunity for technological disruption within the private security sector. While robust, enterprise-grade operational management tools exist globally (such as TrackTik) and generic recruitment/hiring platforms dominate the local market, there is a distinct absence of a localized, integrated ecosystem. Existing solutions either fail to address the specific operational workflows of security firms or lack a client-facing service booking interface. SGCMP is strategically positioned to fill this gap by combining a security service marketplace serving both individual and organizational Clients with a SaaS-based workforce and operational management platform for Security Companies.

By introducing SGCMP, we capture a powerful, dual-sided market opportunity:

- **For Clients (The Demand Side):** The platform acts as a digital trust broker by reducing the lack of transparency and inefficiency associated with traditional security service hiring methods through a centralized marketplace. Clients ranging from individual homeowners to large event organizers can search for and compare published Security Companies, review their profiles, service information, pricing, and previous customer feedback, and submit service booking requests to suitable providers. Security Companies can then review these requests and provide quotations as part of the service engagement process. After a service contract has been completed, Clients can

submit ratings and reviews, helping improve marketplace transparency, provide useful information for evaluating service quality, and build confidence among prospective Clients.

- **For Security Companies (The Supply Side):** SGCMP serves as an accessible, out-of-the-box catalyst for digital transformation. Most small to medium-sized security firms lack the capital to develop proprietary internal software. By utilizing a Software-as-a-Service (SaaS) subscription model, SGCMP provides these companies with modern management tools at a fraction of the cost.
- **Sustainable Revenue Model:** For the platform itself, this dual-sided approach guarantees a robust monetization strategy. The SaaS model ensures predictable, recurring subscription revenue from security companies, while the marketplace opens potential avenues for transaction-based commissions or premium listing fees, creating a highly scalable and sustainable business model.

5. Software Product Vision

SGCMP is envisioned not merely as a standalone utility, but as a comprehensive Web-based digital ecosystem dedicated to modernizing the private security service industry. The platform aims to bridge the communication and operational gap between clients and security companies by combining a marketplace for service hiring with an internal SaaS-based management system for security agencies.

● Pillar 1: Marketplace Connection — The Customer Hub

A transparent and convenient marketplace designed to simplify the process of finding and engaging security services. Clients can search for published Security Companies, discover nearby providers using their current browser location, filter providers by service type and price, view company profiles and previous customer reviews, and directly compare two Security Companies based on key criteria such as service price and rating. Clients can also submit service booking requests, receive quotations, track booking progress, confirm service contracts, and evaluate service quality after contract completion.

● Pillar 2: SaaS Subscription Ecosystem — The Scalable Business Framework

A subscription-based business model designed to provide Security Companies with dedicated digital workspaces for managing their internal operations. Companies can select suitable SaaS subscription plans, with operational capacity and resource limits determined by the selected plan. This approach enables companies to adopt the platform according to their organizational needs while supporting scalable workforce and operational growth.

● Pillar 3: Company Operations — The Management Center

A back-office management environment designed to reduce manual administrative and operational workload for Security Companies. The platform supports Guard profile and qualification management, workforce performance monitoring, customer booking and quotation processing, service contract management, recurring shift scheduling, Guard assignment, attendance monitoring, and operational performance tracking. For unexpected urgent situations near or during a shift, an authorized Coordinator can dispatch a replacement Guard directly. When a Guard becomes aware in advance that they may be unable to attend an upcoming shift due to illness, family matters, or other unforeseen personal circumstances, they can submit a shift change request for the Coordinator to review and approve or reject. Together, these capabilities provide Company Admin and Coordinator with centralized tools for managing day-to-day security service operations.

● **Pillar 4: Service Communication and Coordination — The Support Channel**

A communication and service quality layer that supports interaction between authenticated Clients and Security Companies. The platform provides web-based real-time messaging for service-related communication and coordination. Clients can also submit operational service incident and complaint reports concerning Guard attendance or conduct, while Coordinator can review and manage these reports through tracked processing statuses. Emergency crisis response, emergency-service integration, and automatic emergency escalation are outside the platform scope and must be handled through external emergency channels or company hotlines.

6. Project Scope & Limitations

6.1 Major Features

- **FE-01: Account & Role Management:** Manage accounts and system access permissions based on roles including Platform Admin, Company Admin, Coordinator, Security Guard, Customer, and Guest.
- **FE-02: Security Company & SaaS Package Management:** Manage information of security companies participating in the platform, individual management workspaces for each company, SaaS package usage status, and subscription payment history.
- **FE-03: Security Company Profile & Service Management:** Allow companies to manage their profiles, introductory information, operating areas, types of services provided, and pricing for display on the Marketplace.
- **FE-04: Search, Filter & Compare Security Companies:** Allow customers to search for companies by geographical area, filter by service type and price, view detailed profiles, and compare services between companies.
- **FE-05: Service Request Management:** Allow customers to send security guard hiring requests to companies; allow companies to receive, review, and process service requests.
- **FE-06: Service Contract Management:** Manage contracts between customers and security companies, support both parties in confirming/agreeing to sign online, and track contract status.
- **FE-07: Security Guard Profile Management:** Allow companies to create, update, and manage Security Guard profiles along with necessary information for personnel coordination activities.
- **FE-08: Shift Management & Staff Coordination:** Allow companies/Coordinators to create work shifts, assign Security Guards to customer locations, and provide assigned staff information to customers.
- **FE-09: Shift Execution Confirmation & Management:** Support Security Guards in confirming assigned shifts and recording the status of shift execution/completion as a basis for tracking activities and evaluating performance.
- **FE-10: Shift Exchange & Staff Replacement Management:** Support Security Guards in requesting shift exchanges during unexpected events and support companies/Coordinators in handling replacements and adjusting personnel for work shifts.
- **FE-11: Security Guard Performance Tracking:** Aggregate and track the work execution information of Security Guards based on shift data to support companies in evaluating staff performance.
- **FE-12: Service Rating & Customer Communication:** Allow customers to review/rate the company's quality after using the service and support direct chat between customers and the security company.

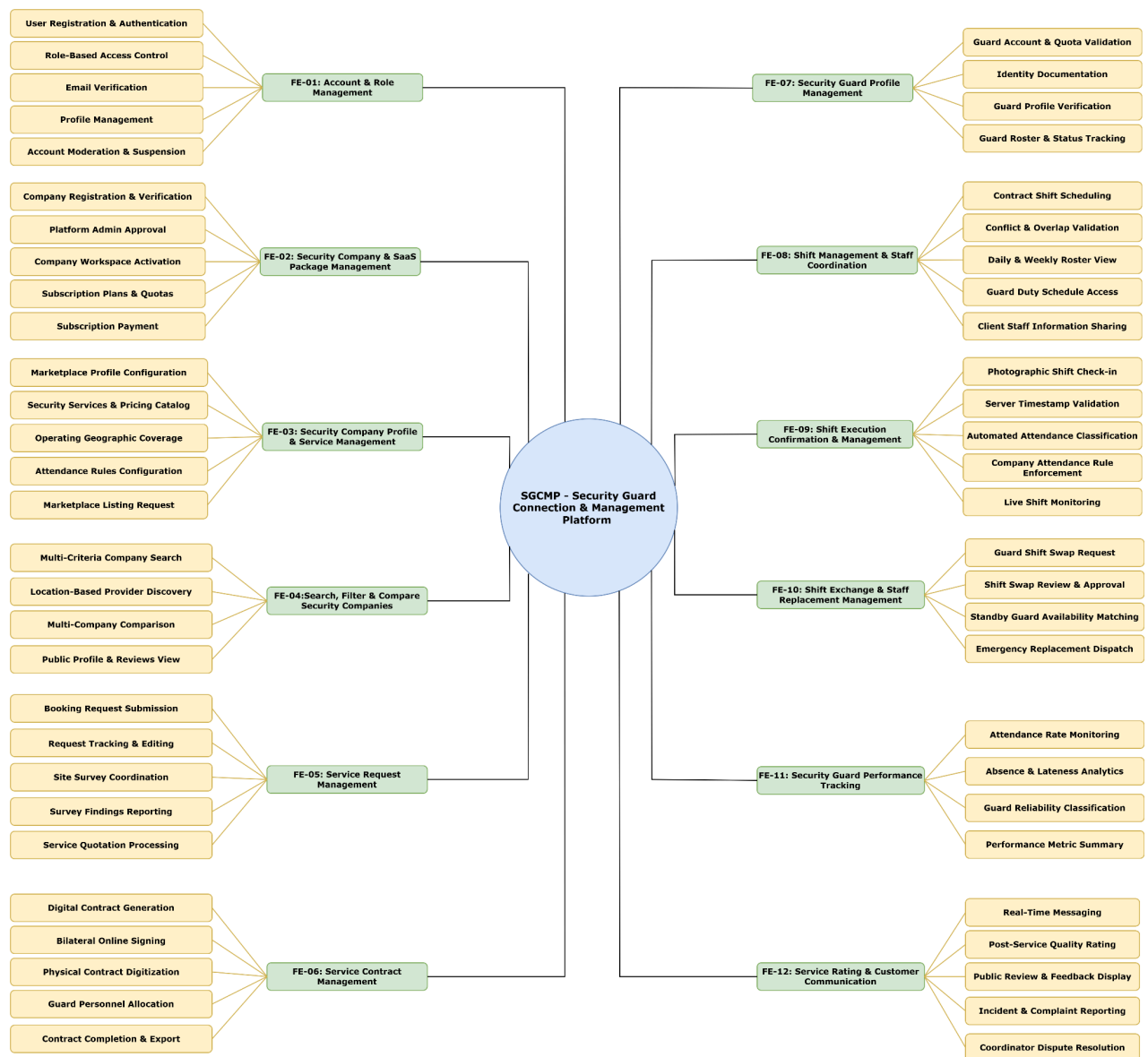


Figure 9. Major Feature

6.2 Limitations & Exclusions

Due to time and project scope constraints, the SGCMP system does not include the following functions:

- **LI-01: Security Service Payment:** The system does not process payments, deposits, refunds, reconciliations, or commission splitting for service contracts between customers and security companies. Payments on the system only apply to **the security company's SaaS packages**.
- **LI-02: Location Verification During Shifts:** The system does not use GPS, geofencing, or positioning technology to verify the location of security guards during check-in. Shift confirmation is based solely on recorded time and image information.
- **LI-03: Specialized Equipment Integration:** The system does not integrate with specialized equipment such as biometric time attendance machines, RFID/NFC, patrol devices, or access control systems.
- **LI-04: Emergency Situation Handling:** The system does not support emergency response coordination, connections with emergency agencies/services, or automatic handling of serious security situations.

- **LI-05: Mobile Application:** The system does not develop native mobile applications for iOS or Android; users access it via a responsive Web Application.
- **LI-06: Payroll and Accounting:** The system does not support calculating payroll, taxes, insurance, payslips, or in-depth corporate accounting operations.
- **LI-07: External Certificate Verification:** The system does not connect with external agencies or systems to verify the authenticity of citizen ID cards (CCCD), licenses, professional certificates, or health documents.
- **LI-08: Digital Signatures and Contract Authentication:** The system does not provide third-party digital signatures, digital certificates, or electronic authentication services; the contract function only records the **online confirmation/agreement between the customer and the security company**.